

Wave 1 Directional Best Hyperparameter Search Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

- Campaign Name: wave1_directional_best_hyperparameter_search_campaign_2026_05_11_19_41_11
- Artifact Commit: 5cf40ebe2f3625f6e202237d4ed06265f5b9659c
- Closeout Timestamp: 2026-05-17T11:55:57
- Bounded Grid Surfaces: 6
- Optuna Surfaces: 9
- Total Surfaces: 15
- Best Hyperparameters Available: True
- Native Python Artifacts Verified: True
- ONNX Exports Verified: True

The closeout combines the 6 bounded-grid tree and harmonic_regression surfaces with the 9 persisted Optuna neural surfaces.

Search Completion

Phase	Surface Count	Completion Evidence
bounded_grid	6	campaign_leaderboard.yaml and family registries
optuna	9	best_trial.yaml , study_summary.yaml , and trial result snapshots

Best Hyperparameters

Family	Scope	Engine	BestRun	Best Hyperparameters	Canonical Family Best?
tree	global	bounded_grid	te_hist_gbr_tabular_global_grid_depth10_lr008_leaf10	model.max_depth=10; model.learning_rate=0.08; model.min_samples_leaf=10; dataset.point_stride=5	True
tree_fw	forward	bounded_grid	te_hist_gbr_tabular_fw_grid_depth6_lr008_leaf10	model.max_depth=6; model.learning_rate=0.08; model.min_samples_leaf=1	True

Family	Scope	Engine	BestRun	Best Hyperparameters	Canonical Family Best?
				0; dataset.poin t_stride=5	
tree_bw	backward	bounded_grid	te_hist_ gbr_tabular_ Bw_grid_ depth6_ lr008_ leaf10	model.max_d epth=6; model.learn ing_rate=0.08 ; model.min_sa mples_leaf=1 0; dataset.poin t_stride=5	True
harmonic_regr ession	global	bounded_grid	te_ harmonic_ order12_ linear_ conditioned_ recovery_ global_grid_ order12_ lr00005_ stride5	model.harmo nic_order=12 ; training.lea rning_rate=0 .0005; dataset.poin t_stride=5	True
harmonic_regr ession_fw	forward	bounded_grid	te_ harmonic_ order12_ linear_ conditioned_ recovery_Fw_ grid_order8_ lr00005_ stride5	model.harmo nic_order=8; training.lea rning_rate=0 .0005; dataset.poin t_stride=5	True
harmonic_regr ession_bw	backward	bounded_grid	te_ harmonic_ order12_ linear_ conditioned_ recovery_Bw_ grid_order8_ lr0002_ stride5	model.harmo nic_order=8; training.lea rning_rate=0 .002; dataset.poin t_stride=5	True
feedforward	global	optuna	te_ feedforward_ stride1_ high_ compute_ long_remote_ global_ optuna_ t0001	training.le arning_rate= 0.0004891706 203866443; training.wei ght_decay=3. 095566460242 3687e-06; dataset.curv e_batch_size =8; model.dropou t_probabilit y=0.1; model.hidden _size=[256, 128, 64]	False
feedforward_f w	forward	optuna	te_ feedforward_ stride1_ high_ compute_ long_remote_ Fw_optuna_ t0005	training.le arning_rate= 0.0003178260 5138591587; training.wei ght_decay=1. 037973382929 3252e-06; dataset.curv e_batch_size =8; model.dropou	False

Family	Scope	Engine	BestRun	Best Hyperparameters	Canonical Family Best?
				t_probabil y=0.0; model.hidden _size=[256, 128, 64]	
feedforward_b w	backward	optuna	te_ feedforward_ stride1_ high_ compute_ long_remote_ Bw_optuna_ t0000	training.le arning_rate= 0.0007106591 851092234; training.wei ght_decay=0. 000368086081 48776104; dataset.curv e_batch_size =2; model.dropou t_probabilit y=0.1; model.hidden _size=[256, 256, 256, 128]	False
periodic_mlp	global	optuna	te_ periodic_ mlp_h04_ standard_ global_ optuna_ t0006	training.le arning_rate= 0.0003109682 534027558; training.wei ght_decay=3. 520862591283 0525e-06; dataset.curv e_batch_size =4; model.dropou t_probabilit y=0.0; model.hidden _size=[128, 128, 64]; model.harmon ic_order=6	False
periodic_mlp_ fw	forward	optuna	te_ periodic_ mlp_h04_ standard_Fw_ optuna_ t0001	training.le arning_rate= 0.0006044543 365247595; training.wei ght_decay=2. 607965659809 584e-05; dataset.curv e_batch_size =8; model.dropou t_probabilit y=0.15; model.hidden _size=[128, 128, 64]; model.harmon ic_order=6	False
periodic_mlp_ bw	backward	optuna	te_ periodic_ mlp_h04_ standard_Bw_ optuna_ t0006	training.le arning_rate= 0.0003109682 534027558; training.wei ght_decay=3. 520862591283 0525e-06; dataset.curv e_batch_size =4;	True

Family	Scope	Engine	BestRun	Best Hyperparameters	Canonical Family Best?
				<pre> model.dropout_probability=0.0; model.hidden_size=[128, 128, 64]; model.harmonic_order=6 </pre>	
residual_harmonic_mlp	global	optuna	<pre> te_residual_h12_deep_joint_wave1_global_optuna_t0011 </pre>	<pre> training.learning_rate=0.00043116294467215643; training.weight_decay=0.00011347122039083106; dataset.curve_batch_size=4; model.residual_dropout_probability=0.0; model.residual_hidden_size=[128, 128, 64]; model.harmonic_order=8 </pre>	False
residual_harmonic_mlp_fw	forward	optuna	<pre> te_residual_h12_deep_joint_wave1_Fw_optuna_t0011 </pre>	<pre> training.learning_rate=0.0006427381768091817; training.weight_decay=1.0093958069558592e-05; dataset.curve_batch_size=8; model.residual_dropout_probability=0.0; model.residual_hidden_size=[192, 128, 64]; model.harmonic_order=12 </pre>	False
residual_harmonic_mlp_bw	backward	optuna	<pre> te_residual_h12_deep_joint_wave1_Bw_optuna_t0005 </pre>	<pre> training.learning_rate=0.002643977767420395; training.weight_decay=0.0003294325864613329; dataset.curve_batch_size=2; model.residual_dropout_probability=0.1; model.residual_hidden_size=[192, 128, 64]; model.harmonic_order=8 </pre>	False

HPO Winner Ranking

Rank	Family	Scope	Engine	Val MAE [deg]	Test MAE [deg]	Test RMSE [deg]
1	tree_fw	forward	bounded_grid	0.002677	0.002743	0.003409
2	tree	global	bounded_grid	0.002655	0.002782	0.003520
3	tree_bw	backward	bounded_grid	0.002681	0.002954	0.003749
4	harmonic_regression_fw	forward	bounded_grid	0.002848	0.003101	0.003527
5	periodic_mlp_bw	backward	optuna	0.002907	0.003233	0.003792
6	periodic_mlp	global	optuna	0.002964	0.003233	0.003733
7	feedforward_bw	backward	optuna	0.002875	0.003276	0.003767
8	feedforward_fw	forward	optuna	0.002746	0.003287	0.003911
9	periodic_mlp_fw	forward	optuna	0.002751	0.003294	0.003899
10	residual_harmonic_mlp_fw	forward	optuna	0.002759	0.003354	0.003995
11	residual_harmonic_mlp	global	optuna	0.002868	0.003428	0.003928
12	feedforward	global	optuna	0.002958	0.003446	0.004158
13	residual_harmonic_mlp_bw	backward	optuna	0.002930	0.003454	0.003918
14	harmonic_regression_bw	backward	bounded_grid	0.003638	0.003494	0.004081
15	harmonic_regression	global	bounded_grid	0.017025	0.020774	0.022412

Artifact Verification

Family	Scope	Native Format	Native Status	ONNX Status	ONNX Size [B]
tree	global	.pk1	ok	ok	313782
tree_fw	forward	.pk1	ok	ok	290566
tree_bw	backward	.pk1	ok	ok	290566
harmonic_regression	global	.ckpt	ok	ok	6647
harmonic_regression_fw	forward	.ckpt	ok	ok	5334
harmonic_regression_bw	backward	.ckpt	ok	ok	5334
feedforward	global	.ckpt	ok	ok	184441
feedforward_fw	forward	.ckpt	ok	ok	184410
feedforward_bw	backward	.ckpt	ok	ok	684512

Family	Scope	Native Format	Native Status	ONNX Status	ONNX Size [B]
periodic_mlp	global	.ckpt	ok	ok	122741
periodic_mlp_fw	forward	.ckpt	ok	ok	122772
periodic_mlp_bw	backward	.ckpt	ok	ok	122741
residual_harmonic_mlp	global	.ckpt	ok	ok	118553
residual_harmonic_mlp_fw	forward	.ckpt	ok	ok	154521
residual_harmonic_mlp_bw	backward	.ckpt	ok	ok	153400

Export Archive

- Export Root: `models\exported`
- Export Inventory: `models\exported\wave1_directional_hpo_export_inventory.yaml`
- `tree` surfaces use `.pkl` Python artifacts.
- `harmonic_regression` and neural surfaces use `.ckpt` Python artifacts.
- Every surface has a refreshed `onnx/model.onnx` export in `models/exported/`.

Registry Note

HPO winners and canonical family-best winners are related but not always identical. The canonical registries use the repository selection policy based on held-out test metrics, while `Optuna` selects the best trial by `val_mae`. This closeout records the HPO winners and explicitly flags whether each one is also the current canonical family-best entry.